STAT2004 – Tutorial Week 12

Exercises 1 and 2 will be covered in the tutorial. Please try Exercises 3 and 4 at home after this week's lectures.

Exercise 1 (Grades and majors[1])

Grades from a statistics class, classified by the students' majors, are tabulated below.

Grade	Psychology	Biology	Other	Total
A	8	15	13	36
B	14	19	15	48
C	15	4	7	26
D–F	3	1	4	8
Total	40	39	39	118

Is there an association between grade and major?

(a) Write down the null and alternative hypotheses, either in words or using symbols.

(b) As a first attempt at answering this question, let's try using Pearson's χ^2 test of association. We can do this by hand, or we can automate the analysis in `R` using:

```
# input all grades as a single vector
grades = c(8,15,13,14,19,15,15,4,7,3,1,4)

# convert grades to a 4x3 table
table1 = matrix(grades, byrow=T, nrow = 4)

# carry out Pearson's chisq test of association
chisq.test(table1)
```

Why did `R` give us a warning message? Let's look at the calculated expected counts:

```
chisq.test(table1)$expected
```

When there are fewer than 5 counts (expected and/or observed) in some cells, then the asymptotic χ^2 approximation to Pearson's test statistic is no longer valid. We can use **Fisher's exact test** instead.

(c) Write out the logical argument behind Fisher's exact test in the context of this scenario.

(d) Carry out Fisher's exact test on these data using the following command:

```
chisq.test(table1, simulate.p.value = T)
```

and write a short conclusion summarizing your findings.

[1] Adapted from Problem 13.8 of *Mathematical Statistics & Data Analysis* (Rice, 1995)

Exercise 2 (Weight gain and protein diet[2])

To investigate weight gain and diet quality, 15 male weaning rats were randomly assigned to one of three protein diets. Their weight gains (in grams) after one week are tabulated below:

Diet	weight gain (g)					group mean
Low	3.89	3.87	3.26	2.70	3.82	
Medium	8.54	9.32	8.76	9.30	10.45	
High	20.39	24.22	30.91	22.78	26.33	

In this exercise, we manually demonstrate the ANOVA sum-of-squares decomposition.

(a) Write down the one-way ANOVA model corresponding to this experiment.

(b) State the null and alternative hypotheses that is associated with this ANOVA.

(c) For these data, compute

(i) the sample means $\overline{Y}_{j\bullet}$ of each group $j \in \{\text{Low}, \text{Medium}, \text{High}\}$;

(ii) the overall sample mean $\overline{Y}_{\bullet\bullet}$ of all observations.

(d) For these data, compute

(i) the between-groups sum-of-squares $\mathrm{SS}_{\mathrm{diet}} = 5\sum_j(\overline{Y}_{j\bullet} - \overline{Y}_{\bullet\bullet})^2$;

(ii) the within-groups residual sum-of-squares $\mathrm{SS}_{\mathrm{residual}} = \sum_{j,i}(Y_{ji} - \overline{Y}_{j\bullet})^2$;

(iii) the sum $\mathrm{SS}_{\mathrm{diet}}+\mathrm{SS}_{\mathrm{residual}}$ of the group and residual sums-of-squares ;

(iii) the total sum-of-squares $\mathrm{SS}_{\mathrm{total}} = \sum_{j,i}(Y_{ji} - \overline{Y}_{\bullet\bullet})^2$.

(e) What are the degrees-of-freedom associated with $\mathrm{SS}_{\mathrm{diet}}$, $\mathrm{SS}_{\mathrm{residual}}$, and $\mathrm{SS}_{\mathrm{total}}$?

(f) Using your calculations from parts (d)–(e), complete the following ANOVA table:

Source	df	SS	MS	F	P(F)
Diet					
Residual					
Total					

What do you conclude about weight gain and protein diet levels from this table?

(g) Verify your calculations by carrying out the ANOVA in `R` using:

```
# input all weight gains as a single vector
weight.gain = c( 3.89, 3.87, 3.26, 2.70, 3.82,
                 8.54, 9.32, 8.76, 9.30,10.45,
                20.39,24.22,30.91,22.78,26.33)

# input diets as a single vector, in the same order as the obs
diet = c(rep("Low",5), rep("Medium",5),rep("High",5))

# compute the ANOVA decomposition on the data
anova(lm(weight.gain ~ diet))
# why use "lm" here?
# we will see that ANOVA is a special case of a linear model (lm)
```

[2] Adapted from Exercise 11.21 of *Statistical Inference* (Casella & Berger, 2002)

Exercise 3 (Two-group ANOVA versus two-sample t test)

Consider the following one-way ANOVA model,

$$Y_{ji} = \mu_j + \epsilon_{ji} \ , \quad j = 1, 2, \ i = 1, 2, \ldots, r_j \ ,$$

where $\epsilon_{ji} \overset{\text{iid}}{\sim} N(0, \sigma^2)$ across both indices j and i. Note that the number of replications are not necessarily equal in the two groups.

(a) Write out the expressions for the between-groups sum-of-squares SS_{group} and the within-group residual sum-of-squares $\text{SS}_{\text{residual}}$.

(b) Explain why $df_{\text{group}} = 1$ for the between-group sum-of-squares, and $df_{\text{residual}} = r_1 + r_2 - 2$ for the residual sum-of-squares.

(c) Show that the F-ratio given by

$$F = \frac{\text{MS}_{\text{group}}}{\text{MS}_{\text{residual}}} := \frac{\text{SS}_{\text{group}}/df_{\text{group}}}{\text{SS}_{\text{residual}}/df_{\text{residual}}}$$

is equivalent to T^2, where T is the usual two-sample t-test statistic given by

$$T = \frac{\overline{Y}_{1\bullet} - \overline{Y}_{2\bullet}}{S_{\text{pooled}}\sqrt{1/n_1 + 1/n_2}}$$

(d) State the distributions of both F and T under the null hypothesis $H_0 : \mu_1 = \mu_2$.

(e) Using parts (c) and (d), what can you say about the relationship between a t_d distribution with d degrees-of-freedom and an $\mathcal{F}_{1,d}$ distribution with 1 and d degrees-of-freedom?

(f) In the case where we only have two groups, what is an advantage of a two-sample t-test over an ANOVA?

Exercise 4 (Working with different ANOVA parametrisations)

In class, we introduced the one-way ANOVA model using the **mean parametrization** as

$$Y_{ji} = \mu_j + \epsilon_{ji} \ , \quad j = 1, 2, ..., J, \ i = 1, 2, \ldots, r_j \ ,$$

where $\epsilon_{ji} \overset{\text{iid}}{\sim} N(0, \sigma^2)$ across both indices j and i. Under this parametrization, the model parameters μ_j are interpreted as the population group means of each group j.

Let's consider two other equivalent parametrizations of the *same* model. For simplicity, we consider the balanced case with $r_1 = r_2 = \ldots = r_J = r$ for all groups.

The "sum constraint" parametrization:

$$Y_{ji} = \mu + \alpha_j + \epsilon_{ji} \ , \quad j = 1, 2, ..., J, \ i = 1, 2, \ldots, r \ , \text{ with } \textstyle\sum_{j=1}^{J} \alpha_j = 0 \ .$$

(a) Give an interpretation of the model parameters μ, $\alpha_1, \alpha_2, \ldots, \alpha_J$ under the sum constraint parametrization.

(b) Show that the MLEs of the model parameters under the sum parametrization are

$$\begin{aligned} \hat{\mu} &= \overline{Y}_{\bullet\bullet} \ , \\ \hat{\alpha}_j &= \overline{Y}_{j\bullet} - \overline{Y}_{\bullet\bullet} \ , \quad j = 1, 2, \ldots, J \ , \\ \hat{\sigma}^2 &= \frac{1}{Jr} \sum_{j=1}^{J} \sum_{i=1}^{r} \left(Y_{ji} - \overline{Y}_{j\bullet} \right)^2 \ . \end{aligned}$$

(c) What are the fitted means $\hat{\mu} + \hat{\alpha}_j$ for each group j under the sum parametrization?

The "contrast" parametrization:

$$Y_{ji} = \mu + \alpha_j + \epsilon_{ji} \ , \quad j = 1, 2, ..., J, \ i = 1, 2, \ldots, r \ , \text{ with } \alpha_1 \equiv 0 \ .$$

(d) Give an interpretation of the model parameters $\mu, \alpha_2, \ldots, \alpha_J$ under the contrast parametrization.

(e) Show that the MLEs of the model parameters under the contrast parametrization are

$$\begin{aligned} \hat{\mu} &= \overline{Y}_{1\bullet} \ , \\ \hat{\alpha}_j &= \overline{Y}_{j\bullet} - \overline{Y}_{1\bullet} \ , \quad j = 2, \ldots, J \ , \\ \hat{\sigma}^2 &= \frac{1}{Jr} \sum_{j=1}^{J} \sum_{i=1}^{r} \left(Y_{ji} - \overline{Y}_{j\bullet} \right)^2 \ . \end{aligned}$$

(f) what are the fitted means $\hat{\mu} + \hat{\alpha}_j$ for each group j under the contrast parametrization?

By comparing your results from parts (c) and (f), and also the expressions for $\hat{\sigma}^2$ under both parametrizations, convince yourself that these two parametrizations are fitting the *exact same* model to the data.